

ARCHITECTURE

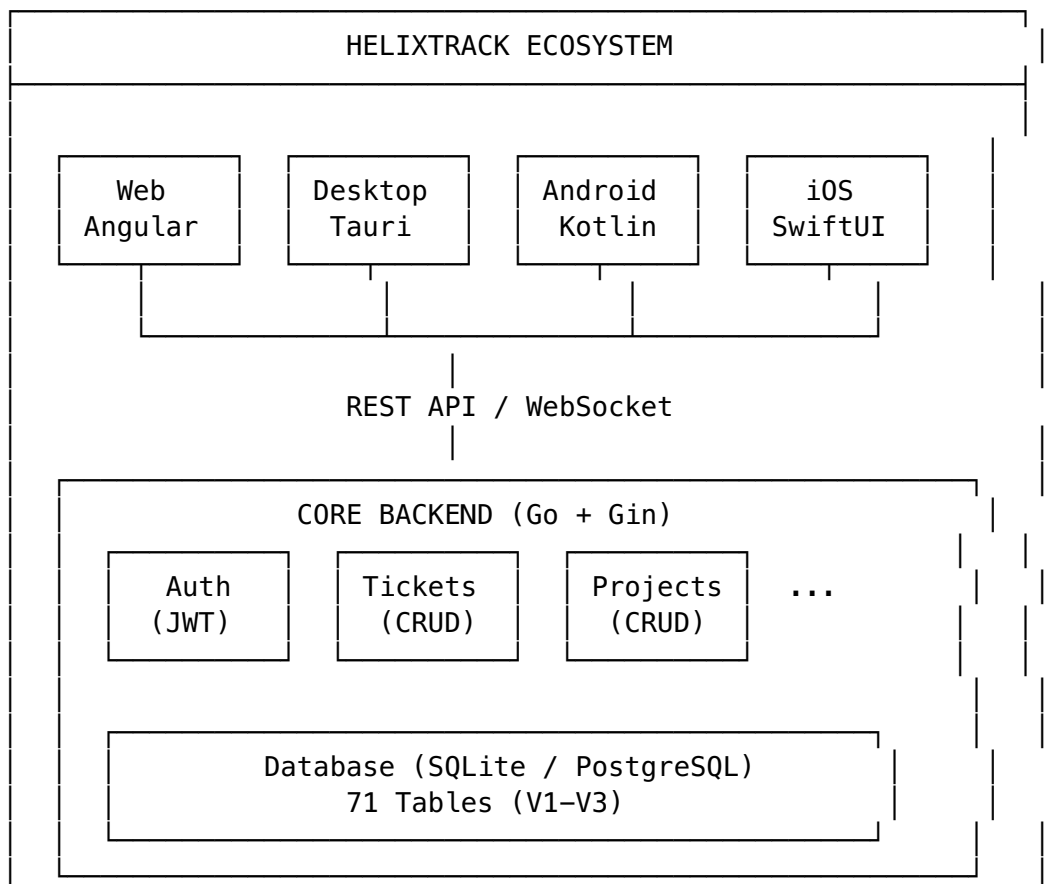
HelixTrack Architecture Overview

What is HelixTrack?

HelixTrack is a comprehensive, modern, open-source JIRA alternative for the free world. It's a multi-platform project management and issue tracking system with a microservices architecture.

Repository: [git@github.com:Helix-Track/Everything.git](https://github.com/Helix-Track/Everything.git) **Status:** V1, V2, V3
Production Ready

System Architecture



Component Inventory

Core Backend (Go)

- **Location:** core/Application/
- **Framework:** Gin Gonic
- **Database:** SQLite (dev) / PostgreSQL (prod)
- **Tables:** 71 (V1-V3)
- **Auth:** JWT + RBAC
- **Features:** REST API, WebSocket, File attachments, Localization
- **Status:** Production Ready

Web Client (Angular 19)

- **Location:** web_client/
- **Framework:** Angular 19 + Angular Material
- **Theme:** Lime Green #BCE63B, Teal #7AA590, Mint #B2E3C2
- **Features:**
 - Dashboard
 - Boards (Kanban)
 - Tickets (CRUD, workflows, relationships)
 - Chat
 - Documents
 - Projects
 - Teams
 - Users
 - Settings
 - Reports
 - Localization management
- **Testing:** Cypress, Playwright, AI QA

Desktop Client (Tauri + Angular)

- **Location:** desktop_client/
- **Framework:** Tauri + Angular
- **Platforms:** macOS, Windows, Linux
- **Features:** Native menus, system tray, window management

Android Client (Kotlin)

- **Location:** android_client/
- **Framework:** Kotlin + Jetpack Compose
- **SDK:** compileSdk 35, minSdk 29
- **DI:** Hilt
- **Navigation:** Navigation component

iOS Client (SwiftUI)

- **Location:** `ios_client/`
 - **Framework:** SwiftUI
 - **Dependencies:** GRDB, Lottie, Markdown UI
 - **Platforms:** iOS 16+, macOS 13+
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Submodules (50+)

HelixTrack incorporates 50+ submodules from vasic-digital and HelixDevelopment organizations:

Core Services

- `auth` — Authentication service
- `database` — Database abstraction
- `storage` — File storage
- `messaging` — Messaging service
- `event_bus` — Event system
- `cache` — Caching layer

AI/ML Services

- `llm_orchestrator` — LLM orchestration
- `llm_provider` — LLM provider abstraction
- `llms_verifier` — LLM verification
- `vision_engine` — Vision AI
- `embeddings` — Embedding service
- `rag` — RAG pipeline

Infrastructure

- `containers` — Docker/container management
- `security` — Security tools
- `observability` — Monitoring
- `optimization` — Performance optimization

Development Tools

- `helix_qa` — QA automation
 - `challenges` — Test challenges
 - `docs_chain` — Documentation sync
 - `doc_processor` — Document processing
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Database Schema (71 Tables)

V1 Core (57 tables)

- `project` — Projects/workspaces
- `ticket` — Issues/tasks
- `user` — Users
- `team` — Teams
- `board` — Kanban boards
- `comment` — Comments
- `attachment` — File attachments
- ... and 50 more

V2 Extensions (14 tables)

- Enhanced features
- Workflow management
- Reporting

V3 Advanced

- AI integration
 - Advanced analytics
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API Design

REST Endpoints

- `/api/v1/projects` — Project CRUD
- `/api/v1/tickets` — Ticket CRUD
- `/api/v1/users` — User management
- `/api/v1/teams` — Team management
- `/api/v1/boards` — Board management
- `/do` — Generic action endpoint

WebSocket

- Real-time updates
 - Chat
 - Notifications
-

Theme System

Brand Colors

- **Primary:** Lime Green #BCE63B
- **Secondary:** Teal #7AA590
- **Accent:** Mint #B2E3C2

CSS Custom Properties

```
--helixtrack-primary: #BCE63B;  
--helixtrack-secondary: #7AA590;  
--helixtrack-accent: #B2E3C2;  
--helixtrack-background: #FFFFFF;  
--helixtrack-surface: #F8F9FA;  
--helixtrack-text-primary: #1A1A1A;
```

Integration Points

CodeGraph

- Local SQLite code-knowledge-graph
- MCP server for CLI agents
- Indexes all source code

OpenDesign

- UI design system
- Design tokens and themes
- MCP server for design generation

Media Validator

- Validates media artifacts
 - OCR/vision verification
 - Evidence-based testing
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Cross-references

- [HelixTrack README](#)
- [Core CLAUDE.md](#)
- [Web Client Guide](#)
- [Desktop Client Guide](#)

Sources verified 2026-06-22: /Volumes/T7/Projects/helix_track/ (sibling HelixTrack source tree) , <https://github.com/Helix-Track/Everything>

REPO-STATE-DERIVED (per §11.4.99 the sources are the cross-referenced HelixTrack source tree + upstream umbrella, following the docs/ARCHITECTURE.md precedent). Cross-referenced on 2026-06-22: - **Client roster CONFIRMED** against the upstream umbrella repo github.com/Helix-Track/Everything (Web / Desktop / Android / iOS / Aurora OS / Harmony OS / Core) AND the present sibling tree (helix_track/{core,web_client,desktop_client,android_client,aurora_os_client,auth,Database} all present). - **Microservices/AI components CONFIRMED** by the umbrella's submodule set (Auth, Cache, storage, VectorDB, LLMOrchestrator, LLMProvider, embeddings, RAG). - **Negative finding:** the umbrella README describes itself as a “wrapper / monorepo umbrella repo” and does NOT explicitly state “JIRA alternative for the free world” — that framing in this doc comes from the project's own component docs, not the umbrella landing page. The `git@github.com:Helix-Track/Everything.git` remote in the header is correct.